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Unit – 4 : Session – 5 : SLO – 1 | Multiple Bus Organization — Fill in the Blanks

■ I. Fill in the Blanks

1. address bus — carries memory addresses from the processor to memory.
2. data bus — transfers data between the processor and I/O devices.
3. Harvard — separate buses for instructions and data define the Harvard architecture.
4. local (backside/cache) bus — connects the CPU to high-speed cache memory.
5. contention (bus bottlenecks) — multiple buses allow simultaneous transfers, reducing contention.
6. Bus A: first source operand to the ALU; Bus B: second source operand (Bus C carries the ALU result back to the registers).
7. control bus — transmits control signals between components.
8. performance (throughput) — parallelism across buses improves overall speed over a single-bus system.
9. bus arbiter — coordinates traffic and grants access among devices/buses.
10. crossbar (point-to-point interconnect) — modern processors use high-speed crossbar/point-to-point buses to link multiple cores to the shared cache.